

Spec: Half-Day Leave

Status

Not implemented

Problem

The current system only supports whole-day leave requests. Employees cannot apply for a half-day (e.g., a morning medical appointment, or leaving early on the first day of a trip), and HR cannot record half-day absences. This means half-days are either rounded up to a full day (unfair to the employee's balance) or not recorded at all.

Scope

This spec covers the minimal change needed to support half-day leave end-to-end:

- Requesting a half-day on the first and/or last day of any leave application (single or multi-day)
- Deducting 0.5 days per marked half-day from the balance
- Displaying half-days correctly in the calendar and approval views

It does **not** cover quarter-days, hours-based tracking, or partial attendance (clocking in late/early).

Design Decisions

When is a half-day applicable?

A half-day can be applied to the **first day**, the **last day**, or both, of any leave request -- including multi-day requests. There is no restriction requiring `startDate === endDate`.

Examples:

- Employee leaves Wednesday afternoon -> marks Wednesday (start) as **PM half-day**
- Employee returns Thursday morning (works PM) -> marks Thursday (end) as **AM half-day**
- Single-day appointment -> marks that day (start) as **AM** or **PM**

Which half?

For each marked day, the employee selects **AM** or **PM**. This is informational for the manager/HR and does not affect the deduction amount (always 0.5 per marked day).

Balance deduction

- Each marked half-day deducts **0.5 working days** from pending balance.
- A request can deduct 0.5 (one half-day), 1.0 (two half-days, or a full day), up to N days.
- Base deduction: `countWorkingDays(startDate, endDate)`, then subtract 0.5 per marked half-day.
- `formatLeaveDays()` already supports decimals -- no display changes needed.

Database Changes

``leave_requests` table -- add two columns`

```
ALTER TABLE leave_requests ADD COLUMN start_half_day TEXT CHECK (start_half_day IN ('AM', 'PM')) DEFAULT NULL;  
ALTER TABLE leave_requests ADD COLUMN end_half_day TEXT CHECK (end_half_day IN ('AM', 'PM')) DEFAULT NULL;
```

- **start_half_day** -- non-null means the first day is a half-day; value indicates which portion
- **end_half_day** -- non-null means the last day is a half-day; value indicates which portion
- For a single-day request, only **start_half_day** is used; **end_half_day** must be null

Update **shared/schema.ts**:

```
startHalfDay: text("start_half_day"), // 'AM' | 'PM' | null  
endHalfDay: text("end_half_day"), // 'AM' | 'PM' | null
```

No changes to **leave_balances** -- it already stores decimals (REAL type).

Backend Changes

Deduction helper -- `server/routes.ts`

Replace the old **isHalfDay ? 0.5 : countWorkingDays(...)** pattern with:

```
function calcRequestedDays(  
  workingDays: number,  
  startHalfDay: string | null,  
  endHalfDay: string | null,  
  isSingleDay: boolean,  
) : number {  
  let days = workingDays;  
  if (startHalfDay) days -= 0.5;  
  if (endHalfDay && !isSingleDay) days -= 0.5;  
  return Math.max(days, 0);  
}
```

Leave creation handler (`POST /api/leave-requests`, ~line 1510)

```
const isSingleDay = validatedData.startDate === validatedData.endDate;  
const workingDays = await countWorkingDays(validatedData.startDate, validatedData.endDate, religion);  
const requestedDays = calcRequestedDays(  
  workingDays,  
  validatedData.startHalfDay ?? null,  
  validatedData.endHalfDay ?? null,  
  isSingleDay,  
);
```

Validation: if **endHalfDay** is set and **startDate === endDate**, reject with 400 -- only **startHalfDay** is valid for single-day requests.

`decrementPending()` (~line 121)

```
const isSingleDay = request.startDate === request.endDate;
const workingDays = await countWorkingDays(request.startDate, request.endDate, religion);
const days = calcRequestedDays(workingDays, request.startHalfDay, request.endHalfDay, isSingleDay);
```

`settlePastApprovedLeave()` (~line 159)

Same pattern -- read **startHalfDay** and **endHalfDay** from the request record and pass through **calcRequestedDays()**.

Schema validation (Zod)

```
startHalfDay: z.enum(['AM', 'PM']).optional().nullable(),
endHalfDay: z.enum(['AM', 'PM']).optional().nullable(),
```

Frontend Changes

`LeaveRequest.tsx`

Below the date picker, always show two optional toggles:

- **"Start as half-day"** -- with AM/PM selector when enabled
- **"End as half-day"** -- with AM/PM selector when enabled; hidden when **startDate === endDate**

The working days preview label reflects the deduction: **countWorkingDays - 0.5 per half-day**.

The balance check uses the same adjusted value.

Submit **startHalfDay** and **endHalfDay** in the request body (null when not enabled).

```
const isSingleDay = dateRange.from?.toDateString() === dateRange.to?.toDateString();

const requestedDays = (() => {
  let days = countWorkingDays(dateRange.from, dateRange.to);
  if (startHalfDay) days -= 0.5;
  if (endHalfDay && !isSingleDay) days -= 0.5;
  return Math.max(days, 0);
})();
```

Calendar views

Half-day events should display with a suffix label ("AM" or "PM") or a half-height block. Pass **startHalfDay** and **endHalfDay** into the calendar event renderer. For multi-day spans, the visual split only applies to the first/last block of the event.

Admin review dialog (`LeaveRequestsSection.tsx`)

Show half-day info next to the working days field:

```
Working Days: 2.5 (start: PM half-day)
Working Days: 2.0 (start: PM half-day, end: AM half-day)
```

Edge Cases

Scenario	Handling
endHalfDay set on a single-day request	Reject with 400 -- only startHalfDay is valid when startDate === endDate
Both half-days marked but only one working day in range	calcRequestedDays floors at 0 via Math.max
Employee applies AM + PM on the same day as separate requests	System doesn't detect overlap; addressed in a future overlap-detection spec
Selected half-day falls on a public holiday	0.5 deducted but the day contributes 0 working days. Warn the employee before
Sick leave half-day, 2-day cert threshold	A single half-day (0.5) is below 2 -- no med cert required

Out of Scope

- Hourly leave tracking
- Half-day attendance clocking changes
- Overlap detection between two half-day requests on the same calendar day